This is a Guide for the literature review to go over for us before we contribute towards the actual paper. (will remove this after the literature review is finished).

What a literature review is

A literature review is a structured summary and critique of existing research that shows what's known, what's missing, and how your project fits in.

Why it's needed (the point)

Establishes context: defines Medicare fraud, typical data, and ML approaches.

Shows gaps: label scarcity, extreme class imbalance, drift, fairness/privacy.

Guides design choices: models, features, metrics, and baselines you should use.

Positions contribution: what your system adds beyond prior work.

What it should include (for your AI Medicare fraud project)

Scope & definitions: fraud vs. waste/abuse; Medicare-specific nuances.

Search strategy: databases (IEEE Xplore, ACM DL, PubMed), years covered, keywords (e.g., Medicare fraud detection, claims anomaly detection, imbalance, graph fraud, Isolation Forest, XGBoost, autoencoder).

Datasets & labeling: claim/provider-level data, public/synthetic options, proxy labels, sampling; HIPAA/privacy constraints.

Feature engineering: claim, provider, temporal, network/graph features; aggregation windows.

Methods: supervised (LR/RF/XGBoost), unsupervised/anomaly (IF, LOF, autoencoders), semi-supervised, graph methods.

Evaluation: precision/recall, PR-AUC (more informative than ROC-AUC under imbalance), cost-sensitive metrics, alert volume; external validity.

Operational constraints: interpretability, latency, human-in-the-loop review, drift monitoring, fairness, compliance.

Synthesis & gaps: what works, where results disagree, and concrete gaps your system targets.

Positioning statement: 3–5 lines on how your approach addresses a specific gap (e.g., provider-level anomaly detection with cost-aware triage + drift checks).

Format & structure (use this)

Length: ~1,200–2,000 words for a course paper; expand if the target journal expects more.

Headings (suggested):

Introduction & Scope

Search Strategy & Inclusion Criteria

Data & Labeling in Medicare Fraud

Feature Engineering Approaches

Methods for Fraud Detection (Supervised, Unsupervised, Graph)

Evaluation Metrics & Practical Constraints

Synthesis of Findings & Gaps

How Our Work Advances the Field

Limitations of the Literature & Future Directions

Style: match target journal (APA or IEEE are common in CS/health). Use consistent in-text citations and a references list.

Tables/figures (highly recommended):

Table 1: Summary of prior studies (data, method, metrics, key limits).

Figure 1: Taxonomy of approaches (supervised vs. anomaly vs. graph).

Table 2: Metrics used across papers and why PR-AUC/precision@k matter here.

Quality signals: compare like-to-like (same metrics), avoid cherry-picking, note reproducibility limits.

Mini-outline tailored to your project (fill-in ready)

Intro & Scope (1–2 ¶): Why Medicare fraud matters; problem framing.

Search Strategy (1 ¶): Databases, years, keywords, inclusion/exclusion.

Data & Labels (2–3 ¶): Claim/provider levels, imbalance, proxies, privacy.

Features (2 ¶): Hand-crafted vs. learned, temporal and network features.

Methods (3–4 ¶): Supervised (RF/XGB), anomaly (IF/AE), graph; pros/cons.

Evaluation (2 ¶): PR-AUC, precision@top-K, reviewer workload; cost-aware.

Synthesis & Gaps (2): What works; where evidence is thin; open problems.

Our Positioning (1 ¶): What your system contributes (e.g., cost-aware anomaly triage with drift monitoring + interpretable features).

Limitations & Future (1 ¶): Data access, generalizability, fairness.

Ultra-short template (you can paste into your doc)

Paragraph 1–2: Problem significance + scope.

Paragraph 3: How you searched and selected papers.

Paragraphs 4–8: What others did (data → features → methods → metrics), with a comparison table.

Paragraph 9: Synthesis—patterns, contradictions, gaps.

Paragraph 10: Your project's niche and why it matters.

References: Consistent style (APA or IEEE).

Deliverables checklist

1. 1–2 paragraph intro & scope
2. 3–5 sentence search strategy
3. Comparative table of 8–15 key papers
4. Taxonomy figure (optional but strong)
5. Clear gap statement + your positioning
6. Consistent citations & references